

LifeGuard AI — System Architecture Diagram

Autonomous Background Consumer Defense Daemon • AWS Agents for Humans Hackathon 2026

AWS STRANDS AGENTS SDK

BEDROCK AGENTCORE RUNTIME

1. AMBIENT INGESTION

Runs 24/7 silently in background

Filesystem Daemon Watcher

Monitors: `ambient_inbox/`

- Polling interval: 120s
- PDF & CSV Auto-Parsing
- Line-Item Drift Extraction

Inbound Email Webhook

`protect+human@lifeguard.ai`

- Auto-forwarded invoices
- Zero user configuration
- Ingests statements < 15ms

Orbital LifeOps Radar

18 Household Sentry Channels

- Spotify (\$11.99) -> Silent Pass
- Netflix (\$15.49) -> Silent Pass
- ConEd Power (\$78.20) -> Pass
- Apple iCloud (\$2.99) -> Pass
- Municipal Water -> Silent Pass
- Amazon Prime -> Silent Pass

95% SILENT PASS RATE

- Comcast (+\$35) -> ANOMALY
- Planet Fitness -> ANOMALY
- Quest Hospital -> ANOMALY
- Breville Recall -> ANOMALY

THREAT INTERCEPTED

2. STRANDS AGENTS MULTI-AGENT REASONING

strands-agents v1.55.1 • Deterministic Typed Python Tools

Node 1: IngestionDaemon

Silent Event Extraction & Token Normalizer

- Runs in 12ms • Parses contract baseline vs current charge • Zero alert noise

Node 2: Strands AuditAgent

Tool: `@tool audit_bill_drift()` & `check_safety_and_warranty()`

- Calculates monthly variance Δ and annualized compounding leakage: $12 \times (C_{curr} - C_{base})$
- Detects stealth fees (\$7.50 Network Modernization Fee) & expiring warranties

Node 3: Strands PolicyAgent

Tool: `@tool lookup_consumer_rights()`

- Federal CFR Statutory Indexing Engine:
 - FCC 47 C.F.R. § 8.1 (Broadband Transparency) • FTC 16 C.F.R. Part 425 (Click-to-Cancel)
 - CMS 45 C.F.R. § 149.410 (No Surprises Act) • CPSC 15 U.S.C. § 2064 (Safety Recalls)

Node 4: Strands ResolverAgent

Tool: `@tool draft_action_resolution()`

- Synthesizes assertive, legally binding executive dispute briefs & formal demand letters
- Identifies executive care emails, postal addresses & regulatory CCs (`consumercomplaints@fc`)

Amazon Bedrock Foundation Model Layer

- Primary: Anthropic Claude 3.5 Sonnet • Fallback: Amazon Nova Pro • `agentcore.yaml` specific

3. HITL GOVERNANCE & EXECUTION

Zero-Trust Security Gate • Never Dispatches Without Human Release

Node 5: Zero-Trust HITL Gatekeeper

Autonomous execution strictly stops at financial/contractual boundary

- Mandatory threshold: > \$10.00 drift or legal cancellation triggers human lock
- Assembles structured Turkey Decision Card (`dec_4a1c21a`)
- Emits Server-Sent Events (SSE) telemetry to real-time client UI

Surfaced Decision Card & Action Center

Presented to Human Principal on Screen

- Comparative Bill Diff Inspector: Contract Baseline (\$50) vs Invoice (\$84.99)
- Fee Creep Progress Meter (+70% drift)
- Statutory Violation Badges (FCC 47 C.F.R. § 8.1)
- 1-Click Action Gate: [Authorize & Dispatch] or [Dismiss / Modify]

HUMAN AUTHORIZATION REQUIRED

Verified Household Financial Recovery Ledger

Dispatch & Permanent Audit Trail

- Dispatches official legal brief to executive care & regulatory portals
- Generates immutable confirmation reference (e.g. `CONF-XFIN98214`)
- Credits verified recovered dollars to Household Ledger
- Benchmark Suite Total Defended: \$2,339.71

CONFIRMED HOUSEHOLD SAVINGS: \$2,339.71